Onboarding APIs into Tyk

Onboarding an API into Tyk means making Tyk aware of your API so it can manage, protect, and monitor it. This flow is designed to help developers ensure their APIs are production-ready and meet organizational requirements.

Define your API

Start by telling Tyk about your API.

You specify:

- The listen path (what path Tyk should expose)
- The target URL (where requests are forwarded)
- Whether the API uses HTTP or HTTPS

This is the foundation, Tyk needs to know where to send traffic and what to call it.

Secure the API

Once registered, the next concern is access control.

Tyk supports various authentication mechanisms:

- API Keys (most basic)
- OAuth 2.0 or JWT (token-based access)
- mTLS (certificate-based)

This ensures only authorized clients can use your API. It is optional but recommended.

Access Control Rules Using Policies

Tyk uses policies to control how your API is used:

- Rate limiting (requests per minute/hour/day)
- Quotas (total requests allowed)
- Access permissions (what endpoints and methods are allowed)

Policies are reusable and can be assigned to different users or applications. It is optional but recommended.

For JWT authentication, policies are mandatory to apply access control rules to the API request.

Customize API Behavior with Middleware

Tyk has a wide variety of Out Of The Box plugins for extending API behavior:

- Add custom headers
- Modify request/response payloads
- Validate input data
- Block/transform requests

This gives you flexibility without changing backend services.

Monitoring and Observability

Tyk provides analytics out of the box using the Dashboard GUI

You can track:

- Number of requests
- Status codes
- Latency
- Errors

For deeper visibility, you can integrate with tools like Prometheus, Grafana, or OpenTelemetry.

Running a smoke test

Before going live, test your API from within the Tyk Dashboard or using tools like Postman.

- The routing works correctly
- Security is enforced
- Rate limiting behaves as expected

You’ve successfully created, secured, and deployed your first API